

Reachable sets in a thin-layer limit

A two-page summary of a preliminary PhD research proposal

Ethan Knox · Champaign, IL · ethank5149@gmail.com · October 6, 2026

Mathematical core, with the proof: research.ekdynamics.science/proposal.pdf

The problem. The reachable set of an atmospheric entry vehicle, that is, every state it can be driven to under admissible control, is what a safety analysis needs bounded. Computing it exactly means solving a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in the state dimension and capped in practice near four or five states [1]; a rigid-body entry vehicle carries thirteen, and more once ablation and aerothermoelastic deformation are coupled. A polynomial barrier whose coefficients come from an interior-point solver has analytical *form*, but it inherits the solver’s accuracy and is not by itself a certificate. There are therefore two problems to solve, not one: the dimension of the state, and the provenance of the guarantee. The vehicle and its corridor enter only as data, so the bound sought is for hypersonic entry in general.

The analytical difficulty. Entry supplies a small parameter, the ratio $\varepsilon = H_s/R_e$ of the atmospheric scale height to the planetary radius. Measured on the planetary length scale the sensible atmosphere is a layer of thickness ε in which the aerodynamic force is of size ε^{-1} , so a vehicle that dips into it spends a time of order ε there and leaves with its velocity changed by a fixed amount. Call such a dip a *pass*. As $\varepsilon \rightarrow 0$ a pass becomes instantaneous and keeps its effect, and the limiting objects stop seeing it: an occupation measure weighs an interval of time by its length, and a pass has no length in the limit. Since a reachable set is a support and not an integral, what has no mass still counts. The remedy is a decomposition into a low-dimensional *skeleton*, which is what survives the limit; one *bubble* per pass, the pass rescaled to its own width, whose net effect is a jump; and a *residual*, bounded explicitly.

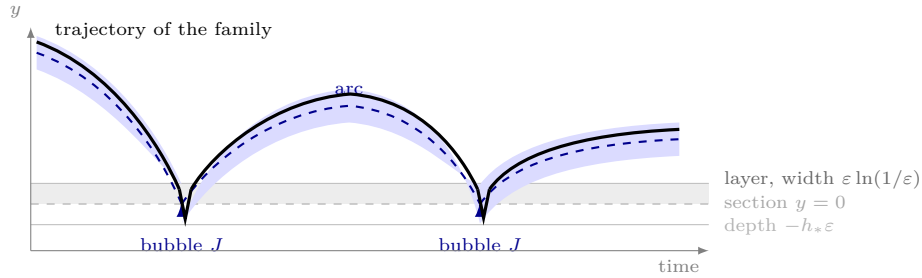


Figure 1: The theorem, as a schematic. A trajectory of the family (solid) descends, rebounds inside the layer in a time of order ε , and climbs again; the limit system (dashed) flows along the arcs and jumps at the section, by the pass map J . The two stay within the residual (shaded). The layer and the depth are drawn far thicker than they are: both shrink with ε while the arcs do not, which is the whole of the difficulty. Nothing here is computed.

A theorem. The question is whether the decomposition is a bound, and the core proves that it is in a model that keeps the concentration and discards everything else. For $\varepsilon \in (0, 1]$ let the state be $(y, v, s) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^d$, with y an altitude, v its rate and s a vector of slow coordinates, and let

$$\dot{y} = v, \quad \dot{v} = a + \rho_\varepsilon(y) L, \quad \dot{s} = F + \rho_\varepsilon(y) \Gamma, \quad \rho_\varepsilon(y) = \varepsilon^{-1} e^{-y/\varepsilon},$$

where a, L, F, Γ depend on s , on a control, and on a selection from a bounded family of admissible fields — so that the system is a differential inclusion and not a single vector field. The factor ε^{-1} is not a device: it is what the atmosphere supplies, since $\int_0^\infty \rho_\varepsilon dy = 1$ for every ε , and without it a pass would contribute $O(\varepsilon)$ and leave no trace in the limit. Assume the field is Lipschitz on a compact corridor, that $a \leq -a_{\min} < 0$ and $L \geq L_{\min} > 0$, that some C^1 function of s decreases at a rate proportional to ρ_ε (in entry this is the energy, since $\dot{\mathcal{E}} = -P \leq 0$ with P the drag power), and that the section $y = \ell_1$ is crossed transversally with a margin v_* . Let $\mathcal{H}$ be the *limit system*: it flows by $\dot{y} = v, \dot{v} = a, \dot{s} = F$, and at $y = 0$ it jumps by a pass map J defined as the reachable set of an inner problem carrying no small parameter. Then there are $\varepsilon_1 > 0$ and a constant C , both explicit in the data, with

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\mathcal{R}_\mathcal{H}([T - \varrho, T + \varrho])}_{\text{skeleton and bubbles}} \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}}, \quad \varrho = C \varepsilon \ln(1/\varepsilon),$$

for all $\varepsilon \leq \varepsilon_1$, where $\mathcal{R}_\varepsilon(T)$ is the reachable set of the family at time T . The inclusion is uniform over the controls, over the initial set and over the whole selection family, which is the sense in which an outer bound has to hold if it is to be a certificate.

Every step of the proof is elementary, and that is deliberate. The dissipation hypothesis integrates once to a *budget*, bounding $\int_0^T \rho_\varepsilon dt$ by a constant independent of ε and of the control. The budget bounds the number of passes,

because each must reverse a velocity of at least v_* ; it bounds the time spent below the section, because the density there is at least ε^{-1} ; and a change of variable from time to delivered impulse makes the inner problem exactly free of ε , with the neglected terms carrying a factor that integrates back to the elapsed time, which is short. Gronwall's inequality assembles the pieces. No compactness argument and no measure appears anywhere in it. We had expected to need the concentration-compactness principle of Lions [2] for the structure, and it remains the natural tool in the general case; but a budget is a stronger thing to have, because it comes with a constant.

Where the measures enter. The theorem hands the occupation-measure program of Lasserre, Henrion and collaborators [3] a different problem. $\mathcal{R}_{\mathcal{H}}$ is a reachability problem in $2 + d$ coordinates with one guard and a set-valued reset, which is the hybrid form for which the relaxation is known [4]: the Liouville equation is a linear constraint on a measure, a measure on the guard carries the mass across each pass, and the dual certificate is a pair of functions non-decreasing along the flow and across the reset. The singular limit is discharged by the theorem, which is elementary; the dimension and the provenance of the guarantee are discharged on $\mathcal{H}$, which is low-dimensional by construction. That division of labour is the shape of the program.

What remains. The hypotheses above are satisfied by entry except for the last, and the failure is quantitative: a pass entered at flight-path angle γ_e enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman variables [5], and for the Earth $\sqrt{\varepsilon} \approx 0.034$ is about two degrees while entry angles are a few degrees. A pass is therefore neither short compared with the arc that follows it nor distinct in scale from the oscillation about the equilibrium glide. Three problems follow.

1. **Capture.** The transversality hypothesis excludes a trajectory whose passes weaken and accumulate, and that is capture into the equilibrium glide. The skeleton must carry the glide as a second flow, with the passage into it inside the limit system rather than in the residual. The budget survives; only the pass count is lost, and it is to be replaced by a threshold on the energy a pass dissipates.
2. **The section moves.** A change of bank moves the glide altitude by a fixed fraction of a scale height however thin the layer, so the section is control-dependent and the amplitude of the oscillation is a coordinate on which the control acts: a bank history in step with the oscillation pumps it, just as one opposed to it damps it. Guidance laws use the latter [6]; a reachable set must allow for the former.
3. **Separation of scales.** The attitude and elastic motions do not concentrate. They are persistent oscillations, as is the oscillation about the glide, and in a profile decomposition such motions belong to the remainder [7]. The tool for them is averaging [8], but their frequencies cross during every pass, the available results give orders and not bounds, and they hold through a resonance only outside a set of small measure, which an outer bound cannot discard. What is needed is an averaging theorem with explicit constants, uniform over the admissible family and valid through resonance.

The third problem is the critical path, and it is the part of the program for which I am looking for an advisor.

- [1] S. Bansal et al. "Hamilton-Jacobi Reachability: A Brief Overview and Recent Advances". In: *56th IEEE Conference on Decision and Control (CDC)*. 2017, pp. 2242–2253.
- [2] P.-L. Lions. "The Concentration-Compactness Principle in the Calculus of Variations. The Locally Compact Case, Part 1". In: *Annales de l'Institut Henri Poincaré, Analyse Non Linéaire* 1.2 (1984), pp. 109–145.
- [3] J.-B. Lasserre et al. "Nonlinear Optimal Control via Occupation Measures and LMI Relaxations". In: *SIAM Journal on Control and Optimization* 47.4 (2008), pp. 1643–1666.
- [4] V. Shia et al. "Convex Computation of the Reachable Set for Controlled Polynomial Hybrid Systems". In: *53rd IEEE Conference on Decision and Control (CDC)*. 2014, pp. 1499–1506.
- [5] D. R. Chapman. *An Approximate Analytical Method for Studying Entry into Planetary Atmospheres*. Tech. rep. Technical Note 4276. National Advisory Committee for Aeronautics, 1958.
- [6] P. Lu. "Entry Guidance: A Unified Method". In: *Journal of Guidance, Control, and Dynamics* 37.3 (2014), pp. 713–728.
- [7] H. Bahouri and P. Gérard. "High Frequency Approximation of Solutions to Critical Nonlinear Wave Equations". In: *American Journal of Mathematics* 121.1 (1999), pp. 131–175.
- [8] J. A. Sanders, F. Verhulst, and J. Murdock. *Averaging Methods in Nonlinear Dynamical Systems*. 2nd ed. Springer, 2007.